

Evonne McArthur, MD, PhD

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EDUCATION & TRAINING

Jun 2023 – Pres.	University of Washington, Department of Medicine <i>Internal Medicine – Medical Genetics Resident Physician, PGY3</i>	Seattle, WA
May 2023	Vanderbilt University School of Medicine <u>M.D.</u> • USMLE Step 1: 258 · USMLE Step 2 CK: 266 · USMLE Step 3: 251	Nashville, TN
May 2022	Vanderbilt University <u>Ph.D. Human Genetics</u> • Advisor: John A. Capra, PhD · Co-advisor: Nancy J. Cox, PhD • Committee: Lea K. Davis, PhD; Douglas Ruderfer, PhD; Alexander Bick, MD, PhD; Emily Hodges, PhD • Dissertation: “The contribution of the 3D genome to gene regulation, human evolution, and disease”	Nashville, TN
Jan 2021 – Mar 2022	University of California, San Francisco <i>Visiting Scholar, Bakar Computational Health Sciences Institute</i>	San Francisco, CA
May 2015	University of North Carolina at Chapel Hill • GPA: 3.95 · Minors: Computer Science & Chemistry · Honors Carolina member	Chapel Hill, NC
Fall 2013	School for International Training, Study Abroad • Program: India – Health and Human Rights	Delhi, India
May 2011	William G. Enloe Magnet High School • International Baccalaureate (IB) Diploma	Raleigh, NC

HONORS & AWARDS

May 2023	Albert Weinstein Prize in Medicine, Vanderbilt Department of Medicine • Recognizing high academic achievement, superior clinical competence, and the qualities of dedication and professionalism that characterize a good physician
Oct 2022	First-place digital medical education innovation submission • Awarded at Digital Education (iMED) 2022: Building Skills, Community and the Future of Medical Education
2021, 2022	Second-place senior human genetics graduate student talk of the academic year
2021	Predoctoral semifinalist, Charles J. Epstein trainee award, American Society for Human Genetics (ASHG)
2020	First-place human genetics graduate student talk of the academic year
2018–2019	Meade Haven Scholarship in Biomedical Sciences
2016–2023	Canby Robinson Society Scholarship • Full-tuition 4-year merit scholarship to Vanderbilt University School of Medicine
2013	Order of the Grail-Valkyries Inductee • Recognized for outstanding character and significant contribution to the intellectual climate at UNC
2013	Phi Beta Kappa Inductee
2013	Phillips Ambassador Scholarship Awardee • Selected to represent UNC as an ambassador to Asia (New Delhi, India)

GRANTS

- 2020 - 2023 **NIH National Research Service Award (NRSA) F30 (F30HG011200)**
- **PI:** Evonne McArthur, Vanderbilt Univ · **Funding IC:** NHGRI
 - **Title:** Quantifying the relationship between 3D genome structure and the genetic architecture of common complex disease
 - **Sponsor:** John A. Capra, PhD · **Co-sponsor:** Nancy J. Cox, PhD

CONSULTING

- Feb 2023 – Dec 2024 **Glass Health**
Associate Managing Editor (Leadership Team)
- Led clinical team and provided informatics support during initial stages of medical technology start-up with the mission of leveraging AI to support clinical reasoning and personalized medical documentation

LEADERSHIP & SERVICE

- 2024 – Pres. **UW Medicine Comfort Care Quilts**
Member
Webmaster
- Quilter, volunteer, and website creator and manager for organization that donates quilts to hospitalized patients during end-of-life care
- 2016 – Pres. **Medical, Pre-medical and pre-MD/PhD student mentorship**
Mentor
- Mentor to UCLA MD/PhD student)
 - Mentor to one UCSF Biological and Medical Informatics graduate student (advised rotation project)
 - Mentor to two Vanderbilt MD/PhD students for scientific presentation skills
 - Mentor to 14 undergraduate students through SNMA (1), SIM program (1), Alpha Epsilon Delta (2), MSTP Summer Research program (3), APSA (4), and informal contacts (3)
 - Selected mentees include:
 - Isabella Lin, PhD – mentee while MD/PhD student in residency application
 - Maggie Benson – mentee while undergrad, subsequently MD/PhD trainee at Kansas Univ.
 - Anirban Chakraborty – mentee while postbacc, subsequently MD/PhD trainee at Weill Cornell
 - Carly Stewart – mentee while undergrad, subsequently MD/PhD trainee at Univ. of Cincinnati
- 2018 – 2019 **Careers in Medicine Clinical Informatics Interest Group**
Co-Founder
- 2019 – 2020 *Co-Chair*
- Co-founded and co-chaired the nation's first AAMC Careers in Medicine Clinical Informatics interest group at Vanderbilt
- 2019 – 2020 **Vanderbilt MSTP volunteer and leadership positions**
Women in Training (WIT) MSTP Committee
- Co-founded, co-chaired, and member of an MSTP committee organized to support women physician-scientist trainees via community, service, and advocacy programming
- 2019 – 2022 *MSTP Admissions Committee Interviewer*
- 2019, 2021 *Mini-MSTP middle school scientific engagement program*
- 2019 – 2023 *MSTP Website Committee*
- 2018 – 2020 *MSTP Student Advisory Council Class Representative*
- 2018 – 2019 **Vanderbilt School of Medicine Student Technology Committee**
Chair
- 2016 – 2023 *Member*
- An official liaison between medical students, administration, and IT leaders regarding all technologies related to Vanderbilt School of Medicine
 - Helped transition EHRs (to Epic) and trained all medical students
- 2016 – 2022 **Shade Tree Clinic (Medical student-run clinic for the uninsured and underserved)**
Monthly Clinical Care Provider and Peer Educator
- 2018 – 2019 *Website Master for 2019 Shade Tree Trot 5K (600+ participants, raising \$17,000+)*

TEACHING & CURRICULUM DESIGN

Feb 2022: Two-part workshop	Manuscript writing for reproducible science in MS word and Latex • https://github.com/CapraLab/lab-manuscript-template
Nov 2021: Workshop	Introduction to version control in Git • https://emcarthur.notion.site/Git-Version-Control-a1315177f4fc44c28bf3a54b5f3a3501
Jun 2021: Workshop	Communicating science through interactive apps with Dash and Plotly in python
May 2020: Five-part course	Introduction to scientific and statistical programming • [1] Introduction to computer programming & R • [2] Introduction to data manipulation • [3] Introduction to plotting & regression • [4] Introduction to machine learning • [5] Introduction to single-cell analysis • https://medschool.vanderbilt.edu/mstp/2020/05/28/mstp-programming-workshop/
Aug 2018, 2019: Orientation session	Clerkship student introduction to medical student technology (Epic training)
Fall 2015: one-month course module	UNC BIOL 528 • TA, lecturer, and curriculum design for a one-month module in an advanced undergraduate systems biology course

PUBLICATIONS

Preprints, prepared manuscripts, & articles in submission

- [1] **McArthur, E.**, Khan, K. R., & Buber, Y. (2025). Novel COL1A1 Variant Associated with Heritable Thoracic Aortic Aneurysm and Connective Tissue Disease. *In preparation*.
- [2] **McArthur, E.**, Nesselrooij, B. P. M., Neufeld-Kaiser, W., Miller, D. E., Storz, S. H. R., Goffena, J., Anderson, Z. B., Else, T., Disteche, C., Handford, C., Dubard-Gault, M. E., van Leeuwen, R. S., & Raskind, W. H. (2025). Recurrent contiguous gene deletions that encompass VHL and SLC6A1 result in a combined neurodevelopmental and tumor syndrome. *In preparation*.
- [3] **McArthur, E.**, Rinker, D. C., Gilbertson, E. N., Fudenberg, G., Pittman, M., Keough, K., Pollard, K. S., & Capra, J. A. (2025). Reconstructing the 3D genome organization of Neanderthals reveals that chromatin folding shaped phenotypic and sequence divergence. *bioRxiv*. [In review at *Cell Genomics*]. <https://doi.org/10.1101/2022.02.07.479462>
- [4] Schuler, B. A., Shirazi, S. M., Sucre, J. M. S., Jasper, E. A., **McArthur, E.**, Ruderfer, D. M., & Bastarache, L. (2025). Phenotypic predictors of genetic testing and diagnosis in critically ill neonates. *In preparation*.
- [5] Wang, B., Miller-Fleming, T. W., Yu, D., Hucks, D., Gantz, E., Johnston, R., Maxwell-Horn, A., Cox, N., Sutcliffe, J., Mathews, C. A., **McArthur, E.**, Hatfield, H., Kabir, D., Giangrande, E. J., Fortgang, R. G., Wang, S. B., Karmacharya, R., Roffman, J. L., Scharf, J. M., ... Davis, L. K. (2025). Development and validation of electronic health record-based ascertainment of obsessive-compulsive disorder cases and controls. *medRxiv*. <https://doi.org/10.1101/2025.08.05.25332874>

Peer-reviewed journal

[†] denotes co-first author

- [1] [†]Gjoni, K., [†]Gunsalus, L. M., [†]Kuang, S., [†]**McArthur, E.**, Pittman, M., Capra, J. A., & Pollard, K. S. (2025). Comparing chromatin contact maps at scale: methods and insights. *Nature Methods*, 22(4), 824–833. <https://doi.org/10.1038/s41592-025-02630-5>
- [2] Strom, N. I., Gerring, Z. F., Galimberti, M., Yu, D., Halvorsen, M. W., Abdellaoui, A., Rodriguez-Fontenla, C., Sealock, J. M., Bigdeli, T., Coleman, J. R., Mahjani, B., ..., **McArthur, E.**, ..., & Mattheisen, M. (2025). Genome-wide analyses identify 30 loci associated with obsessive-compulsive disorder. *Nat. Genet.* <https://doi.org/10.1038/s41588-025-02189-z>
- [3] Tubbs, C., Benton, M. L., **McArthur, E.**, Capra, J. A., & Ruderfer, D. M. (2025). Identifying deleterious noncoding variation through gain and loss of CTCF binding activity. *The American Journal of Human Genetics*, 112(4), 892–902. <https://doi.org/10.1016/j.ajhg.2025.02.009>
- [4] Brand, C. M., Kuang, S., Gilbertson, E. N., **McArthur, E.**, Pollard, K. S., Webster, T. H., & Capra, J. A. (2024). Sequence-Based Machine Learning Reveals 3D Genome Differences between Bonobos and Chimpanzees. *Genome Biology and Evolution*, 16(11). <https://doi.org/10.1093/gbe/evae210>
- [5] Gilbertson, E. N., Brand, C. M., **McArthur, E.**, Rinker, D. C., Kuang, S., Pollard, K. S., & Capra, J. A. (2024). Machine Learning Reveals the Diversity of Human 3D Chromatin Contact Patterns. *Molecular Biology and Evolution*, 41(10). <https://doi.org/10.1093/molbev/msae209>

- [6] **McArthur, E.**, Bastarache, L., & Capra, J. A. (2023). Linking rare and common disease vocabularies by mapping between the human phenotype ontology and phecodes. *JAMIA Open*, 6(1). <https://doi.org/10.1093/jamiaopen/ooad007>
- [7] Scott, T. J., Hansen, T. J., **McArthur, E.**, & Hodges, E. (2023). Cross-tissue patterns of DNA hypomethylation reveal genetically distinct histories of cell development. *BMC Genomics*, 24(1). <https://doi.org/10.1186/s12864-023-09622-9>
- [8] Velazquez-Arcelay, K., Colbran, L. L., **McArthur, E.**, Brand, C. M., Rinker, D. C., Siemann, J. K., McMahon, D. G., & Capra, J. A. (2023). Archaic introgression shaped human circadian traits. *Genome Biology and Evolution*, 15(12). <https://doi.org/10.1093/gbe/evad203>
- [9] Niarchou, M., Gustavson, D. E., Sathirapongsasuti, J. F., Anglada-Tort, M., Eising, E., Bell, E., **McArthur, E.**, Straub, P., Aslibekyan, S., Auton, A., & et al. (2022). Genome-wide association study of Musical Beat Synchronization demonstrates high polygenicity. *Nature Human Behaviour*, 6(9), 1292–1309. <https://doi.org/10.1038/s41562-022-01359-x>
- [10] Quach, W. T., Le, C. H., Clark, M. G., **McArthur, E.**, Ancker, J. S., Gadd, C. S., & Johnson, K. B. (2022). Engaging the next generation of physician-informaticians through early exposure to the field: successes and challenges associated with starting a novel clinical informatics interest group. *Journal of the American Medical Informatics Association*, 30(1), 202–205. <https://doi.org/10.1093/jamia/ocac189>
- [11] **McArthur, E.**, & Capra, J. A. (2021). Topologically associating domain boundaries that are stable across diverse cell types are evolutionarily constrained and enriched for heritability. *The American Journal of Human Genetics*, 108(2), 269–283. <https://doi.org/10.1016/j.ajhg.2021.01.001>
- [12] **McArthur, E.**, Rinker, D. C., & Capra, J. A. (2021). Quantifying the contribution of Neanderthal introgression to the heritability of complex traits. *Nature Communications*, 12(1). <https://doi.org/10.1038/s41467-021-24582-y>
- [13] Patel, D. A., Gopalakrishnan, R., Engelhardt, B. G., **McArthur, E.**, Sengsayadeth, S., Culos, K. A., Byrne, M., Goodman, S., Savani, B. N., Chinratanalab, W., Jagasia, M., Mosse, C. A., Cornell, R. F., & Kassim, A. A. (2020). Minimal residual disease negativity and lenalidomide maintenance therapy are associated with superior survival outcomes in multiple myeloma. *Bone Marrow Transplantation*, 55(6), 1137–1146. <https://doi.org/10.1038/s41409-020-0791-y>
- [14] Rinker, D. C., Simonti, C. N., **McArthur, E.**, Shaw, D., Hodges, E., & Capra, J. A. (2020). Neanderthal introgression reintroduced functional ancestral alleles lost in Eurasian populations. *Nature Ecology & Evolution*, 4(10), 1332–1341. <https://doi.org/10.1038/s41559-020-1261-z>
- [15] Lackey, L., Coria, A., Woods, C., **McArthur, E.**, & Laederach, A. (2018). Allele-specific SHAPE-MaP assessment of the effects of somatic variation and protein binding on mRNA structure. *RNA*, 24(4), 513–528. <https://doi.org/10.1261/rna.064469.117>
- [16] **McArthur, E.**, Bali, S., & Khan, A. A. (2016). Socio-cultural and knowledge-based barriers to tuberculosis diagnosis for women in Bhopal, India. *Indian Journal of Community Medicine*, 41(1), 62. <https://doi.org/10.4103/0970-0218.170990>
- [17] Lackey, L., **McArthur, E.**, & Laederach, A. (2015). Increased Transcript Complexity in Genes Associated with Chronic Obstructive Pulmonary Disease. *PLOS ONE*, 10(10). <https://doi.org/10.1371/journal.pone.0140885>
- [18] Capra, R., Chen, A. T., **McArthur, E.**, & Davis, N. (2013). Searcher actions and strategies in asynchronous collaborative search. *Proc. Am. Soc. Info. Sci. Tech.*, 50(1), 1–10. <https://doi.org/10.1002/meet.14505001077>

PRESENTATIONS

Selected oral presentations

* denotes presenter

- [1] *Schuler, B., **McArthur, E.**, Sucre, J. M., Ruderfer, D., & Bastarache, L. (2023). Phenome-wide association studies in the neonatal intensive care population identify phenotypic features that correlate with genetic diagnosis [Platform talk in session: Genomics in the NICU]. *American Society of Human Genetics (ASHG) Meeting*, Washington, DC.
- [2] *Velazquez-Arcelay, K., Colbran, L. L., **McArthur, E.**, Rinker, D. C., & Capra, J. A. (2021). The influence of Neanderthal introgression on human circadian biology [Platform talk in session: Effects of Evolutionary History and Substructure on Modern Populations]. *American Society of Human Genetics (ASHG) Meeting*, Virtual.
- [3] ***McArthur, E.**, Benton, M. L., Tubbs, C., Ruderfer, D., Cox, N. J., & Capra, J. A. (2021). Common-variant dysregulation of Mendelian disorder genes contributes to complex disease [Platform talk in session: Expanding genetic approaches and applications in pharmaco- and clinical genomics]. *American Society of Human Genetics (ASHG) Meeting*, Virtual.
- [4] ***McArthur, E.** & Capra, J. A. (2021). Quantifying the relationship between 3D genome structure and the genetic architecture of complex traits [Plenary Talk]. *NHGRI Research Training and Career Development Annual Meeting*, Virtual.
- [5] **McArthur, E.**, Rinker, D. C., & *Capra, J. A. (2020). Quantifying the contribution of Neanderthal introgression to the heritability of complex traits [Platform talk in session: Effects of Evolutionary History and Substructure on Modern Populations]. *American Society of Human Genetics (ASHG) Meeting*, Virtual.

- [6] ***McArthur, E.** & Capra, J. A. (2020). Personalized regulatory genomics: Identifying gene dysregulation to solve undiagnosed rare disease cases [Platform talk in Session: Computational approaches for disease diagnosis and variant effect determination]. *American Society of Human Genetics (ASHG) Meeting*, Virtual.

Selected poster presentations

* denotes presenter

- [7] ***McArthur, E.** & Webber, C. J. (2022). Schematify: Developing a web application to visualize medical diagnostic schemas. *Vanderbilt Clinical Informatics Center: InformatiCon 2022*, Nashville, TN.
- [8] ***McArthur, E.**, Rinker, D. C., Gilbertson, E. N., Fudenberg, G., Pittman, M., Keough, K., Pollard, K. S., & Capra, J. A. (2021). Reconstructing the 3D chromatin organization of archaic hominins reveals that genome folding shaped human-Neanderthal phenotypic divergence and introgression. *Society for Molecular Biology and Evolution (SMBE) Annual Meeting*, Virtual.
- [9] *Soda, T., **McArthur, E.**, McGuire, P., Crowley, J., Hucks, D., & Davis, L. K. (2019). Identification, validation, and characterization of obsessive disorder case samples in a large US Biobank. *American Psychiatric Association Annual Meeting*, San Francisco, CA.
- [10] ***McArthur, E.** & Capra, J. A. (2019). Heritability of common complex traits is enriched at topologically associating domain boundary regions. *American Society of Human Genetics (ASHG) Meeting*, Houston, TX.
- [11] *Lackey, L., ***McArthur, E.**, Coria, A., Arend, K., Moorman, N., & Laederach, A. (2015). The biological impacts of RiboSNitches: Linking RNA structural changes with molecular phenotypes. *North Carolina Symposium on RNA Biology XI*, Durham, NC.
- [12] ***McArthur, E.**, Zhang, X., Gamazon, E. R., Sutcliffe, J. S., Cook, E. H., Davis, L. K., & Cox, N. J. (2014). Integration of Copy Number and Exome Sequence Data in a Queryable Database for the Investigation of Autism Spectrum Disorder. *International Meeting for Autism Research (IMFAR)*, Atlanta, GA.

Science & medicine writing or applications

- [13] **McArthur, E.** (2022). Interactive web application: Medical diagnostic schema graph generator. <http://diagnostic-schema.herokuapp.com/>
- [14] **McArthur, E.** (2021a). Behind the paper community science article: Uncovering the legacy of our Neanderthal ancestors. Nature Portfolio Ecology & Evolution Community. <https://natureecoevocommunity.nature.com/posts/uncovering-the-legacy-of-our-neanderthal-ancestors>
- [15] **McArthur, E.** (2021b). Interactive web application: Visualizing the theoretical evolutionary trajectory of trait-associated Neanderthal-introgressed alleles. <https://neanderthal-heritability.herokuapp.com/>
- [16] **McArthur, E.** (2019). Interactive web application: Personalized chemotherapy calendar. <http://emcarthur.github.io/projects/ChemoCalendar>

PROFESSIONAL ORGANIZATIONS

2018 – Pres. American Society of Human Genetics (ASHG)
 2021 – 2024 Society for Molecular Biology and Evolution (SMBE)

SKILLS

Software and Computer Languages

- Python, R, MATLAB, Perl, Java, AWK, Linux Shell Scripting, MySQL, SQLite, Mathematica, MS Office, HTML, CSS, Git, L^AT_EX
- Web authoring tools such as Inkscape, Adobe Flash, Illustrator, Photoshop, and Lightroom

Languages

- Introductory conversational Spanish

SKILLS

Art

- Photography: Cover of Dick's Sporting Good's Magazine and featured in many other international publications and texts, Created a two-month 30 photo exhibit exploring the diversity of India's life and landscape
- Small business ([Etsy shop](#)) owner selling homemade digital art with over 350 sales and \$15,000 revenue

Athletics

- Downhill alpine skier: raced slalom and giant slalom for 2011-2014 UNC Ski Team (US Collegiate Ski Association [UCSCA]) and lifelong ski enthusiast
- 5 half marathons 2019-2021

Crafts

- Computer programming and [web application](#) creation
- Others: sourdough bread and pizza making, crochet, linocut printmaking, basic woodworking (Adirondack chairs)